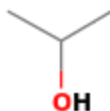


Isopropyl Alcohol

- **Formula:** C₃H₈O
- **Molecular weight:** 60.0950
- **IUPAC Standard InChI:** InChI=1S/C3H8O/c1-3(2)4/h3-4H, 1-2H3  **InChI**
v 1.06
- **IUPAC Standard InChIKey:** KFZMGEQAYNKOFK-UHFFFAOYSA-N 
- **CAS Registry Number:** 67-63-0
- **Chemical structure:**



This structure is also available as a [2d Mol file](#) or as a [computed 3d SD file](#)

[View 3d structure](#) (requires JavaScript / HTML 5)

- **Other names:** 2-Propanol; sec-Propyl Alcohol; Alcojel; Alcosolve 2; Avantin; Avantine; Combi-Schutz; Dimethylcarbinol; Hartosol; Imsol A; Isohol; Isopropanol; Lutosol; Petrohol; Propol; PRO; Takineocol; 1-Methylethyl Alcohol; iso-C₃H₇OH; 2-Hydroxypropane; Propane, 2-hydroxy-; sec-Propanol; Propan-2-ol; i-Propylalkohol; Alcolo; Alcool isopropilico; Alcool isopropylique; Alkolave; Arquad DMCB; iso-Propylalkohol; Isopropyl alcohol, rubbing; IPA; Lavacol; Visco 1152; Alcosolve; i-Propanol; 2-Propyl alcohol; Spectrar; Sterisol hand disinfectant; UN 1219; n-Propan-2-ol; 1-methylethanol; Propanol-2; Virahol; IPS 1
- **Permanent link** for this species. Use this link for bookmarking this species for future reference.
- **Information on this page:**
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- **Other data available:**

- [Gas phase thermochemistry data](#)
- [Phase change data](#)
- [Reaction thermochemistry data](#)
- [Henry's Law data](#)
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 - [Gas Phase Kinetics Database](#)
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- **Options:**
 - [Switch to calorie-based units](#)

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Data compiled as indicated in comments:

ALS - Hussein Y. Afeefy, Joel F. Liebman, and Stephen E. Stein

DH - Eugene S. Domalski and Elizabeth D. Hearing

Quantity	Value	Units	Method	Reference	Comment
$\Delta_f H^\circ_{\text{liquid}}$	-317.0 ± 0.3	kJ/mol	Ccb	Chao and Rossini, 1965	see Rossini, 1934; ALS
$\Delta_f H^\circ_{\text{liquid}}$	-318.2 ± 0.71	kJ/mol	Ccb	Snelson and Skinner, 1961	ALS
$\Delta_f H^\circ_{\text{liquid}}$	-318.7	kJ/mol	Ccb	Parks, Mosley, et al., 1950	see Parks and Moore, 1939; ALS
Quantity	Value	Units	Method	Reference	Comment
$\Delta_c H^\circ_{\text{liquid}}$	-2006.9 ± 0.2	kJ/mol	Ccb	Chao and Rossini, 1965	see Rossini, 1934; Corresponding $\Delta_f H^\circ_{\text{liquid}} = -316.9$ kJ/mol (simple calculation by NIST; no Washburn corrections); ALS
$\Delta_c H^\circ_{\text{liquid}}$	-2005.8 ± 0.4	kJ/mol	Ccb	Snelson and Skinner, 1961	Corresponding $\Delta_f H^\circ_{\text{liquid}} = -318.1$ kJ/mol (simple calculation by NIST; no Washburn corrections); ALS
$\Delta_c H^\circ_{\text{liquid}}$	-2005.1	kJ/mol	Ccb	Parks, Mosley, et al., 1950	see Parks and Moore, 1939; Corresponding $\Delta_f H^\circ_{\text{liquid}} = -318.7$ kJ/mol (simple calculation by NIST; no Washburn corrections); ALS
Quantity	Value	Units	Method	Reference	Comment
S°_{liquid}	180.58	J/mol*K	N/A	Andon, Counsell, et al., 1963	DH
S°_{liquid}	179.9	J/mol*K	N/A	Kelley, 1929	DH
S°_{liquid}	192.9	J/mol*K	N/A	Parks and Kelley, 1928	Extrapolation below 70 K, 43.56 J/mol*K.; DH
S°_{liquid}	190.8	J/mol*K	N/A	Parks and Kelley, 1925	Extrapolation below 90 K, 53.22 J/mol*K.; DH

Constant pressure heat capacity of liquid

$C_{p,liquid}$ (J/mol*K)	Temperature (K)	Reference	Comment
161.2	298.15	Roux, Roberts, et al., 1980	<i>DH</i>
154.75	298.15	Brown and Ziegler, 1979	T = 185 to 304 K. Results as equation only.; <i>DH</i>
165.6	311.6	Griigo'ev, Yanin, et al., 1979	T = 311 to 453 K. p = 0.98 bar.; <i>DH</i>
154.43	298.15	Andon, Counsell, et al., 1963	T = 10 to 330 K.; <i>DH</i>
162.8	298.2	Katayama, 1962	T = 10 to 60°C.; <i>DH</i>
180.3	324.	Swietoslawski and Zielenkiewicz, 1958	Mean value 21 to 81°C.; <i>DH</i>
154.0	298.	Ginnings and Corruccini, 1948	T = 0 to 200°C.; <i>DH</i>
159.99	298.04	Zhdanov, 1945	T = 7 to 41°C. Value is unsmoothed experimental datum.; <i>DH</i>
172.4	303.2	Phillip, 1939	<i>DH</i>
163.6	298.	Trew and Watkins, 1933	<i>DH</i>
149.75	292.84	Kelley, 1929	T = 16 to 298 K. Value is unsmoothed experimental datum.; <i>DH</i>
180.3	298.1	Parks, Kelley, et al., 1929	Extrapolation below 90 K, 42.68 J/mol*K.; <i>DH</i>
151.0	293.1	Parks and Kelley, 1928	T = 71 to 293 K. Value is unsmoothed experimental datum.; <i>DH</i>
152.3	293.1	Parks and Kelley, 1925	T = 71 to 293 K. Value is unsmoothed experimental datum.; <i>DH</i>
169.9	303.	Willams and Daniels, 1924	T = 303 to 323 K. Equation only.; <i>DH</i>

References

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- Symbols used in this document:

$C_{p,\text{liquid}}$	Constant pressure heat capacity of liquid
$S^{\circ}_{\text{liquid}}$	Entropy of liquid at standard conditions
$\Delta_c H^{\circ}_{\text{liquid}}$	Enthalpy of combustion of liquid at standard conditions
$\Delta_f H^{\circ}_{\text{liquid}}$	Enthalpy of formation of liquid at standard conditions

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